Software Project Planning

→ Estimation of cost & development time are key issues during project planning.

→ ~~This~~ Estimation of size, cost & development time of the project can be done before or after finalisation of SRS.

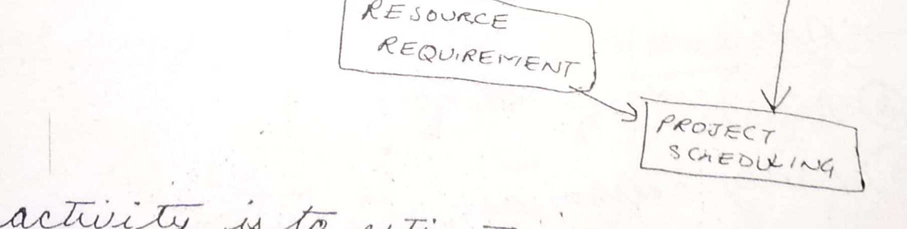
→ Software managers are responsible for planning & scheduling project development.

→ Manager supervise the work to ensure that it is carried out to the required standards.

→ They monitor progress to check that the development is on time & within budget.

→ Good managers cannot guarantee project success but bad manager result in project failure.

Activities during software project planning are:



→ First activity is to estimate the size of the project. Size is key parameter for estimation of other activities.

→ It is an i/p to all costing models for estimation of cost, development time & schedule for the project

→ If size estimation is not reasonable it may have serious impact on the other estimation activities.

→ Resources requirements are estimated on the bas[is]
of cost & development time.
→ Project scheduling may prove to be very useful
for controlling & monitoring the progress of the
project.

(1) SIZE ESTIMATION

(a) Line of Code

→ It is simplest among all metrics available
to estimate project size.
→ project size is estimated ~~using~~ by counting the
number of lines of code.
→ comment and declaration are not considered while counting LOC
→ Defination
↳ A line of code is any line of program text that
is not a comment a blank line, regardless of
the number of statements or fragements of statement
on the line.
↳ This includes all lines containing program
header, declarations & executable & non-executable
statements.

→ Advantage
(a) Simple
(b) Easily recognizable.

Disadvantage.
(1) Lang- dependent
(2) LOC focus on what systems is rather than
what systems does.

(b) FUNCTION COUNT

→ In this measuring the functionality which system provide rather than counting the code or equipment of which systems made of.

→ It measures functionality from the user point of view i.e. on the basis of what the user request & receives in return.

→ It deals with functionality being delivered & not with the lines of code, source modules, files etc.

→ Function point measures functionality from the users point of view, i.e. on the basis of what the user request & receives in return from the system.

FPA is a system decomposed into functional units:

(a) Input: information entering the system.

(b) Outputs: " leaving the system.

(c) Enquiries: requests for instant access to info

(d) Internal logical files: information held within the system

(e) External interface files: inf. held by other system i.e. used by the system being analyzed

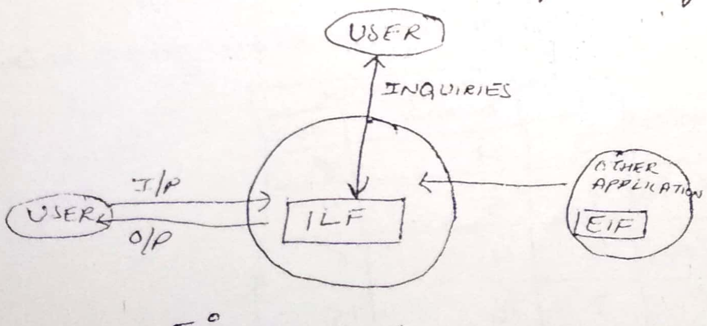


Figure: FPA's functional units.

Functional Units divided into

- Data function
  1. ILF (Internal logical files)
     [logically related data or control info maintained within the system]
  2. EIF (External Interface Files)
     [control information referenced by the system, but maintained within another system.
     This means that EIF counted for one system, may be an ILF in another systems]
- Transactional Function Types
  1. External I/p (EI)
     (control info that comes from outside the system)
  2. External output (EO)
     (control info that ~~comes from~~ to be sent outside the system
  3. External Inquiry (EQ
     [process that is made up of an i/p & o/p combination that result in data retrieval]

Steps to calculate Function Point

(1) Calculate UFP (Unadjusted Function Point (UFP)
where

$$\boxed{UFP = \sum_{i=1}^{5} \sum_{j=1}^{3} Z_{ij} W_{ij}}$$

→ W_{ij} → entry of i^{th} row & j^{th} column
i.e weighting factor as table given below

FUNCTIONAL UNIT	WEIGHTING FACTOR		
	LOW	AVERAGE	HIGH
EXTERNAL I/P (EI)	3	4	6
EXTERNAL O/P (EO)	4	5	7
EXTERNAL (EQ) INQUIRIES	3	4	6
INTERNAL (ILF) LOGICAL FILE	7	10	15
EXTERNAL (EIF) INTERFACE FILES	5	7	10

→ Z_{ij} → complexity of functional unit.

(2) Calculate FP

$$\boxed{FP = UFP \times CAF}$$

CAF is complexity adjustment factor

$$CAF = [.65 + .01 \times \Sigma F_i]$$

F_i (i = 1 to 14) are degree of influence & are based on responses to question noted.

0	1	2	3	4	5
NO influence	Incidental	Moderate	Average	significant	Essential.

(1) Static, Single Variable models

Software Engineering Laboratory of University of marylan (SEI) has established this model.

$$C = aL^b$$

where $C \rightarrow$ cost

a & $b \rightarrow$ constants

$L \rightarrow$ Size, (given in line of code)

$$\boxed{E = 1.4\, L^{0.93}}$$

where $E \rightarrow$ Effort in Person-months

$$\boxed{DOC = 30.4\, L^{0.90}}$$

DOC $\rightarrow$ Documentation, in number of pages.

$$\boxed{D = 4.6\, L^{0.26}}$$

$D \rightarrow$ Duration, in months.

L is predictor

(2) Static, Multivariable Models

=) The model developed by Walston & Felix, It provides a relationship between delivered lines of source code & Effort 'E'

$$E = 5.2\, L^{0.91}$$

$$D = 4.1\, L^{0.36}$$

Productivity

It is expressed in no. of lines of source code per person months

$$\boxed{P = \frac{L}{E}}$$

$$I = \sum_{i=1}^{29} w_i x_i$$

w_i = factor for the weight for the i^{th} variable

$x_i = \{-1, 0, +1\}$

Its value depend on productivity i.e. whether productivity ~~increases~~ decreases, no effect, increases.

(1) Compare the Walston - Felix model with SEL model on a software development expected to involve 8 person-year of effort

(a) calculate the number of lines of source code that can be produced.

(b) calculate the duration of the development

(c) calculate the productivity in LOC/PY

(d) calculate average manning

(1) Given

$E = 8\ PY = 12 \times 8 = 96$ Person-months

(a) $L = ?$

$E = aL^{b}$

$L = (E/a)^{1/b}$

$L(SEL) = (96/1.4)^{1/.93} = 94.26158 = 94.264$ LOC

$L(W.F) = (96/5.2)^{1/.91} = 24.632$ LOC $= 24632$

(b) Duration in months:

$$D(SEL) = 4.6\, L^{0.26}$$

$$= 4.6\,(94.264)^{.26}$$

$$= 15 \text{ months}$$

$$D(W\text{-}F) = 4.1\, L^{0.36}$$

$$= 4.1\,(24.632)^{.36} = 13 \text{ months}$$

(c) Productivity is

$$P(SEL) = \frac{94264}{8} = 11783 \text{ LOC/PY}$$

$$P(W\text{-}F) = \frac{24632}{8} = 3079 \text{ LOC/P-Y}$$

(d) Average manning

$$= E/D$$

$$M(SEL) = \frac{96 \text{ P-M}}{15 \text{ M}} = 6.4 \text{ Person/month}$$

$$M(W\text{-}F) = \frac{96}{13} = 7.4 \text{ Person/month}$$

⇒ Ques:- The size of a s/w product to be developed has been estimated to be 22000 LOC. Predict the manpower cost (effort) by Walston-Felix model & SEL Model.

FORMULAS

	EFFORT	AVG MANNING	DOC	DURATION	PRODUCTIVITY
SEL	$E = 1.4 L^{0.93}$	E/D	$30.4 L^{0.90}$	$D = 4.6 L^{0.26}$	$P = L/E$
WALSTON-FELIX	$E = 5.2 L^{0.91}$	E/D		$D = 4.1 L^{0.36}$	$P = L/E$
UNIT	PERSON-MONTH	PERSON MONTH	NO. OF PAGES	MONTH	LOC/PM

$L \rightarrow$ size (line of code)

COCOMO (CONSTRUCTIVE COST MODEL).

→ It is s/w cost estimation model which include
- → basic
- → Intermediate
- → Detailed sub-Models

→ B.W Boehm's invented this model.

(1) Basic Model

→ Estimation is done in a quick & rough fashion

→ Small to medium projects

→ Modes of Basic Model

Organic	Semi-detached	Embedded
→ Small team	→ Medium size	Large project
→ experienced developers	→ Average experience in similar environment	→ Very little experience
→ Size → from few KLOC to few tens KLOC i.e. small to medium	→ 50 - 300KLOC	→ Over 300KLOC
→ Deadline of the project is Not tight	→ Medium	→ Tight
→ E.g. Payroll project	→ e.g. database systems, editor etc	→ Real project ATM Traffic control

COCOMO equations:

$$\boxed{E = a_b (KLOC)^{b_b}}$$

$$\boxed{D = C_b (E)^{d_b}}$$

E → effort unit Person-Month

D → Development time unit Months

$$\boxed{\text{Average staff size (SS)} = \frac{E}{D}} \text{ Persons}$$

$$\boxed{\text{Productivity (P)} = \frac{KLOC}{E}} \text{ KLOC/PM}$$

COCOMO Coefficients

Project	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

→ # Selection of mode is important, in development time there is no difference, difference is there in effort i.e. manpower

→ effort in ~~Organic~~ Embedded mode is 4 times the organic mode & effort of semi-detached is two times the organic mode

Numerical

Ques: Suppose that a project was estimated to be 400 KLOC. Calculate the effort & development time for each of the three modes i.e. Organic, semi-detached & embedded.

Soln:

$$E = a_b (KLOC)^{b_b}$$
$$D = C_b (E)^{d_b}$$

Size of the project = 400 KLOC

(i) Organic Mode

$$E = 2.4 (400)^{1.05} = 1295.31 \text{ PM}$$
$$D = 2.5 (1295.31)^{0.38} = 38.07 \text{ M}$$

(ii) Semi detached Mode

$$E = 3.0 (400)^{1.12} = 2462.79 \text{ PM}$$
$$D = 2.5 (2462.79)^{.35} = 38.45 \text{ M}$$

(iii) Embedded Mode

$$E = 3.6 (400)^{1.20} = 4772.81 \text{ PM}$$
$$D = 2.5 (4772.8)^{.32} = 38 \text{ M}$$

Ques: A project size of 200 KLOC is to be developed. S/w development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the effort, development time, average staff size & productivity of the project.

Sol: Semi detached mode is used

$$E = 3.0 (200)^{1.12} = 1133.12 \text{ PM}$$
$$D = 2.5 (1133.12)^{.35} = 29.3 \text{ M}$$

Average staff $= \frac{E}{D}$ Persons $= \frac{1133.12}{29.3} = 38.67$ Persons

Productivity $= \frac{KLOC}{E} = \frac{200}{1133.12} = .1765$ KLOC/PM

$\boxed{P = 176 \text{ LOC/PM}}$

Disadvantage of basic model

→ quick & rough estimate
→ Lack of accuracy

2) Intermediate Model

→ Boehm added 15 predicators called cost driver for accuracy.

→ Cost drivers are used to adjust the nominal cost of the project to the actual environment hence cost driver increase the accuracy of the estimate.

Cost driver grouped into 4 categories:

1. Product attributes
   - (a) Required Software Reliability (RELY)
   - (b) Data base size (DATA)
   - (c) Product Complexity (CPLX)
2. Computer attributes
   - (a) Execution time (TIME)
   - (b) Main storage constraints (STOR)
   - (c) Virtual Machine volatility (VIRT)
   - (d) Computer turnaround time (TURN)
3. Personnel attribute
   - (a) Analyst capability (ACAP)
   - (b) Application experience (AEXP)
   - (c) Programmer capability (PCAP)
   - (d) Virtual machine experience (VEXP)
   - (e) Programming lang experience (LEXP)
4. Project attribute
   - (a) Modern Programming practices (MODP)
   - (b) Use of S/w tools (TOOL)
   - (c) Required development schedule (SCED)

-- ost driver is rated for a given project environment.

→ Rating uses a scale very low, low, nominal, high, very high, extra high which describes to what extent the cost driver applies to the project being estimated.

→ EAF (effective adjustment factor)

In this multiplying factors for all 15 cost drivers are multiplied to get the effort adjustment factor (EAF)

EAF range from ·9 to 1·4 [find the value using table]

$$E = a_i (KLOC)^{b_i} * EAF$$

$$D = c_i (E)^{d_i}$$

Value of a_i, b_i, c_i, d_i coefficients :

PROJECT	a_i	b_i	c_i	d_i
ORGANIC	3·2	1·05	2·5	·38
SEMIDETACHED	3·0	1·12	2·5	·35
EMBEDDED	2·8	1·20	2·5	·32

iii: A new project with estimated 400 KLOC embedded system has to be developed. Project manager has a choice of hiring from two pools of developer: Very high capable with very little experience in the programming lang. being used or developers of low quality but a lot of experience with the programming language. what is the impact of hiring all developers from one or the other pool?

Soln: Embedded & intermediate COCOMO

$E = a_i (KLOC)^{d_i}$

$= 2.8 (400)^{1.20} = 3712$ PM

Case I: Developers are very highly capable with very little experience in programming being used.

[value from table
AEXP = .82
LEXP = 1.14 (very little experience in programming)]

$EAF = .82 \times 1.14 = .9348$

$E = 3712 \times .9348 = 3470$ PM

$D = 2.5 (3470)^{.32} = 33.9$ PM

Case II: Developers are of low quality but lot of experience with the programming lang

$EAF = 1.29 \times 0.95 = 1.22$

[AEXP = 1.29 (Very low experience of developer)
LEXP = .95 (High prog. experience)]

$E = 3712 \times 1.22 = 4528$ PM

$D = 2.5 (4528)^{.32} = 36.9$ PM

Case II require more effort & time

3). Detailed COCOMO Model

→ Detailed COCOMO Model introduces two more capabilities:

① Phase - sensitive effort multipliers:

→ Some phases (design, programming, integration / test) are more affected than others by factors defined by cost drivers.

→ It provides a set of phase sensitive effort multiplier for each cost drivers. This helps in determining the manpower allocation for each phase of the project.

② Three - level product hierarchy.

→ Three product levels are defined. These are module, subsystem & system levels.

→ Rating of cost driver is done at appropriate level i.e. level at which it is most susceptible to variation.

Development phases

S/w development is carried out in four successive phase

① Plan / Requirement →

→ In this requirement is analyzed, product plan is set up & full product specification is generated

→ This phase consume

effort → 6% to 8%

development time → 10% to 40%

→ These percentage depend on mode & on size.

(2) Product design (DD%)

→ This phase is concerned with product architecture & specification of the subsystems

→ Effort 16% to 18%

Development 19% to 38%
time

(3) Programming (C%)

~~Effort 24% to 64%~~

Effort 48% to 68%

Development 24% to 64%
time

(4) Integration/Test : (I%)

Effort 16% to 34%

Development 18% to 34%
time

$\boxed{\text{Principle of effort estimate}}$

Size equivalent:

Adjustment factor, A is calculated

→ (1) $\boxed{A = .4DD + .3C + .3I}$

Size equivalent

→ (2) $\boxed{S(\text{equivalent}) = (S \times A)/100}$

S → thousand line of code (KLOC)

→ (3) $E = a(KLOC)^b * EAF$, ~~$D = c(KLOC)^d$~~ $D = c(E)^d$

→ (4) $E_p = \mu_p E$

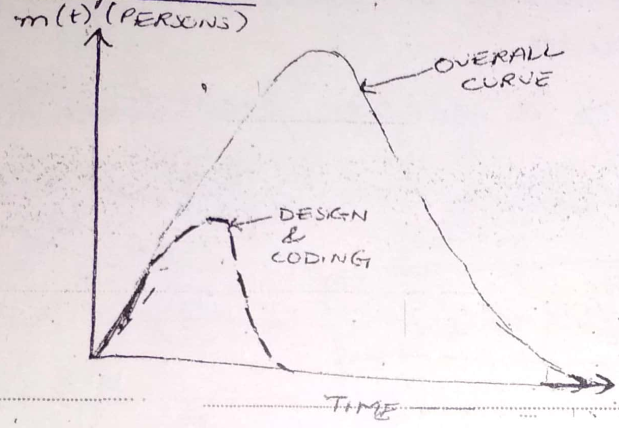
$D_p = \tau_p D$

these calculate phase wise effort & development time

Putnam Resource Allocation Model

→ Rayleigh curve are used for H/w development project

→ Putnam has ~~a~~ used this ~~pro~~ curve for software project. & also for S/w subsystem development.



(1) NORDEN/RAYLEIGH CURVE

→ Norden/Rayleigh equation represents manpower, measured in person per unit time as function of time.

unit → Person-year/year (PY/YR)

Rayleigh curve eqn is.

$$m(t) = \frac{dy}{dt} = 2Kate^{-at^2} \quad \text{—— (1)}$$

where

t → elapsed time

Plays an important Role. ← a → parameter that affects the shape of curve unit → 1/time²

K → area under the curve in interval $[0, \infty]$ OR Effort or Total project cost (person-year)

Integrate eq^n 1

$$\boxed{y(t) = K[1 - e^{-at^2}]} \quad — 2$$

where $y(t) \to$ cumulative manpower.

→ Parameter a

~~Determine~~ the peak

→ parameter 'a' helps in determining the peak manpower

→ Larger value of 'a' earlier peak time occur.

Diff^n eq^n 2

$$\boxed{\frac{d^2y}{dt^2} = 2Kae^{-at^2}[1 - 2at_d^2] = 0} \quad — 3$$

To get Relationship b/w t_d & a

$$\boxed{t_d^2 = \frac{1}{2a}} \quad — 4 \quad \Rightarrow \quad \boxed{a = \frac{1}{2t_d^2}} \quad — 5$$

where

$t_d \to$ Time where Max effort occur
or
total project development time
or
Peak Manning time

Put eq^n 5 in eq^n 2

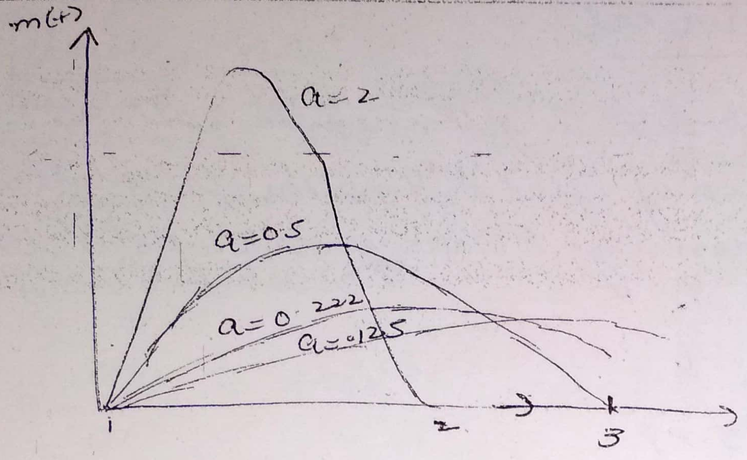
[sustitute 't_d' for 't' in eq 2]

$$E = y(t) = K\left[1 - e^{-\frac{t_d^2}{2t_d^2}}\right]$$

$$= K[1 - e^{-1/2}]$$

$$= K[.3935]$$

$$\boxed{E = .3935K}$$



$m_0 \to$ peak manning i.e. no of persons employed at the peak
unit $\to$ person

$t_d \to$ peak manning time

~~at t~~

peak manning, $m(t_d)$ is obtained, which is denoted by m_0

Put eq^n 5 in eq^n 1

$$m(t) = \cancel{2} K \frac{1}{\cancel{2} t_d^2} t e^{-t^2/2t_d^2}$$

(At time $t = t_d$) Now put $t = t_d$

$$m_0 = m(t_d) = K \frac{1}{t_d^2} \cdot \cancel{t_d} e^{-\cancel{t_d^2}/2\cancel{t_d^2}}$$

$$= \frac{K}{t_d} \cdot e^{-1/2}$$

$$m_0 = m(t_d) = \frac{K}{t_d \sqrt{e}}$$

$$\therefore \boxed{m_0 = \frac{K}{t_d \sqrt{e}}}$$

[Determining number of persons involved at peak time t_d replacing 'a' with '$1/2t_d^2$' in eq (1)]

$$\boxed{\text{avg. rate of team build up} = \frac{m_0}{t_d}}$$

unit $\to$ person/year

Numerical

(1) S/w project is planned to cost 95 PY in a period of 1 year & 9 months. Calculate the peak manning & average rate of S/w team build up?

(1) given:

$K = 95$ PY

$t_d = 1$ year & 9 months → 365 days + 270 days (9×30) = 635/365 = 1.75 yrs

$= 1.75$ yrs.

Peak manning $m_0 = \frac{K}{t_d\sqrt{e}}$

$= \frac{95}{1.75 \times 1.648} = 32.94 \approx 33$ person

Avg rate of S/w team build up

$= m_0/t_d = 33/1.75 = 18.8$ person/year or 15.6 P/M

(2) Consider a large-scale project for which the manpower requirement $K = 600$ PY & development time is 3yr 6 months.

(a) Calculate the peak manning & peak time

(b) What is the manpower cost after 1 year & 2 months?

→ (2) given $t_d = 3$ year & 6 months $= 3.5$ yrs

(a) $m_0 = \frac{K}{t_d\sqrt{e}}$, $m_0 = 600/3.5 \times 1.648 \cong \underline{104}$

(b) $y(t) = K[1 - e^{-at^2}]$, $t = 1$ yr & 2 months $= 1.17$ yrs

$a = \frac{1}{2t_d^2} = \frac{1}{2 \times (3.5)^2} = .041$

$y = 600[1 - e^{-.041(1.17)^2}] = 32.6$ PY

Difficulty Metric

→ The slope of the manpower distribution at start time ($t=0$)
Differentiate Norden Rayleigh Curve i.e.
differentiate foll. eqn

$$m(t) = \frac{dy}{dt} = 2kate^{-at^2}$$

$$m'(t) \frac{d^2y}{dt^2} = 2kae^{-at^2}[1-2at^2]$$

For $t=0$

$$m'(0) = 2kae^{0}[1-2a(0)^2]$$

$$= 2ka$$

Put value of a i.e. $a = \frac{1}{2t_d^2}$

$$= \frac{\cancel{2}K \cdot 1}{\cancel{2}t_d^2} = K/t_d^2$$

$$\boxed{D = \frac{K}{t_d^2}}$$ unit person/year.

Put value of K, as we know $m_0 = \frac{K}{t_d\sqrt{e}} \Rightarrow m_0 t_d\sqrt{e} = K$

$$D = \frac{m_0 \cancel{t_d}\sqrt{e}}{t_d^2} = \frac{m_0\sqrt{e}}{t_d}$$

$$\boxed{D \propto m_0}$$

This relationship shows that a project is more difficult to develop when the manpower demand is high or when the time schedule is short (small t_d)

Manpower buildup

$$\boxed{D_0 = K / t_d^3 \text{ — person/year}^2}$$

Productivity Versus Difficulty

→ Productivity is no of lines of code developed per person-month

→ productivity is proportional to the difficulty

$$P \propto D^{\beta} \quad \text{--- (1)}$$

→ Avg productivity is

$\boxed{P = \text{Line of code produced / Cumulative manpower used to produce code.}}$

$$\boxed{P = S/E} \quad \text{--- (2)}$$

Eq^n (1) is (Putnam analyzed from 50 projects)

$$P = \phi D^{-2/3} \quad \text{--- (3)}$$

Put (2) in (3)

$$\boxed{\frac{S}{E} = \phi D^{-2/3}}$$

$$\boxed{S = \phi D^{-2/3} E} \quad \text{--- (4)}$$

$$= \phi D^{-2/3} (.3935 k) \quad \left[\because E = .3935 k \right]$$

As we know $D = \frac{K}{t_d^2}$, Put this in eq^n (4)

$$S = \phi \left[K / t_d^2 \right]^{-2/3} (.3935 k)$$

$$\cancel{= .3935 \phi K^{1/3} t_d^{-4}}$$

$$= .3935 \phi K^{-2/3} . K . t_d^{-2 \times -2/3}$$

$-2/3+1$

$\frac{-2+3}{3}$

$1/3$

$$\boxed{S = .3935 \phi K^{1/3} . t_d^{4/3}}$$

Replacing $.3935 \phi$ by coefficient 'C' which is technology factor

$$\boxed{S = C K^{1/3} . t_d^{4/3}}$$

OR

$$\boxed{C = S . K^{-1/3} t_d^{-4/3}}$$

Trade-off between Time Versus Cost

As we know $S = C K^{1/3} t_d^{4/3}$

so $\boxed{S/C = K^{1/3} \cdot t_d^{4/3}}$ — (1)

→ Time scale should never be reduced to less than 75% of its intial calculated value because.
If we raise power by 3, then $K t_d^4$ is constant for constant size software. (a) A compression of the development time t_d will produce an increase manpower cost.
(b) If compression is excessive not only would the s/w development cost much more, but also the development would become so difficult that it would increase the risk of being unmanageable.

→ ~~B#~~ Software life cycle Methodology (SLIM) was name given by Putnam to later version of this model.

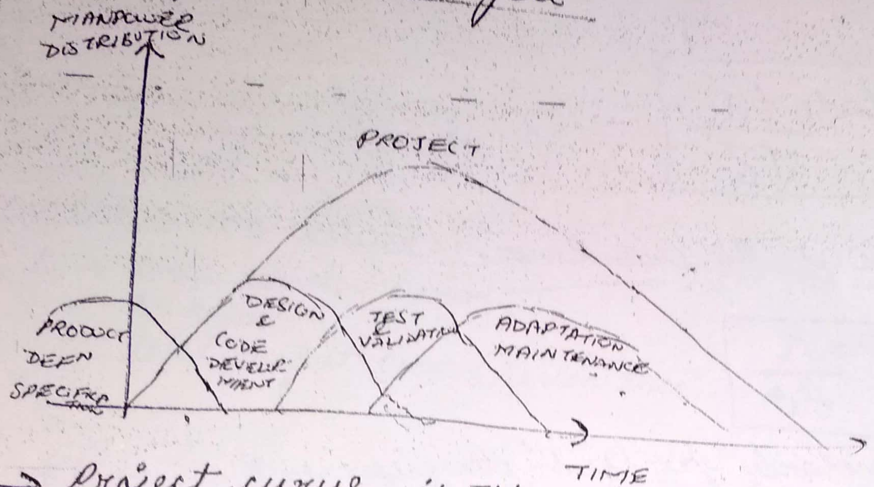
eq^n (1) can be written as

$$\rightarrow \left[S/C\right]^{3} = K \cdot t_d^4$$

$$K = \left[\frac{S}{C}\right]^{3} \cdot \frac{1}{t_d^4}$$

∴ this means $K \propto \frac{1}{t_d^4}$ → effort is inversely proportion to fourth power of development time.

Development Sub-Cycle



→ Project curve is the addition of two curve called development curve & test & validation curve

→ Let $m_d(t)$ → design manning
$y_d(t)$ → cumulative design manpower cost

$m_d(t) = 2K_d\, b t e^{-bt^2}$ $\left[m(t) = 2Kat e^{-at^2} \right]$

$y_d(t) = K_d\left[1 - e^{-bt^2}\right]$ $\left[\because y(t) = K[1 - e^{-at^2}]\right]$

→ Manpower involved in design & coding still completing this activity after t_d

→ Development will be 95% completed by time t_d

$\therefore y_d(t) = K_d\left[1 - e^{-bt_d^2}\right] = .95$

$\frac{y_d(t)}{K_d} = \left[1 - e^{-bt_d^2}\right]$

As we know $a = \frac{1}{2t_d^2}$

Similarly $\boxed{b = \frac{1}{2t_{od}^2}}$

t_{od} → Development time at which development curve exhibits peak manning

$$\boxed{t_{od} = \frac{t_d}{\sqrt{6}}}$$

Now as we know

$$\boxed{D = \frac{K}{t_d^2}}$$

Similarly for sub-development

$$D = \frac{K_d}{t_{od}^2}$$

so $$\frac{K}{t_d^2} = \frac{K_d}{t_{od}^2}$$

Put $t_{od} = t_d/\sqrt{6}$

$$K/t_d^2 = \frac{K_d}{(t_d/\sqrt{6})^2}$$

$$\frac{K}{t_d^2} = \frac{K_d}{\frac{t_d^2}{6}} \Rightarrow \frac{K}{\cancel{t_d^2}} = \frac{6K_d}{\cancel{t_d^2}} \Rightarrow \boxed{K_d = K/6}$$

$$\therefore D_0 = \frac{K}{t_d^3} = \frac{K_d}{\sqrt{6}\, t_{od}^3}$$

Put value of K & td

manpower build-up

$$D_0 = \frac{K}{t_d^3} = \frac{6K_d}{(\sqrt{6}\cdot t_{od})^3} = \frac{\cancel{\sqrt{6}} \times \cancel{\sqrt{6}}\, K_d}{\cancel{\sqrt{6}} \times \cancel{\sqrt{6}} \times \sqrt{6}\, t_{od}^3} \quad [\because 6 = \sqrt{6} \times \sqrt{6}]$$

$$\boxed{D_0 = \frac{K_d}{\sqrt{6}\, t_{od}^3}}$$ persons/year^2

FORMULAS

1. $m(t) = \frac{dy}{dt} = 2Kat e^{-at^2}$

$m(t) \rightarrow$ manpower, unit – PY/Y

$K \rightarrow$ area under curve
or effort
total project cost (Person year)

2. $y(t) = K[1 - e^{-at^2}] \rightarrow \boxed{E = y(t) = .3935\,K}$

$y(t) \rightarrow$ cumulative manpower

$a \rightarrow$ parameter

3. $\boxed{a = \frac{1}{2t_d^2}}$

$t_d \rightarrow$ peak manning time

4. $\boxed{m_0 = \frac{K}{t_d\sqrt{e}}}$ $m_0 \rightarrow$ peak manning
unit persons.

5. $\boxed{\text{avg rate of team build up} = m_0/t_d}$

unit $\rightarrow$ Person/year

6. Difficulty

$\boxed{D = K/t_d^2}$ or $\boxed{D = \frac{m_0\sqrt{e}}{t_d}}$ unit person/year

7. Manpower build up

$\boxed{D_0 = \frac{K}{t_d^3} \text{ person/year}^2}$

(8) Productivity

$P = S/E$

$S \rightarrow$ line of code

$E \rightarrow$ cumulative manpower

~~$S = 39350$~~

$S = C\, K^{1/3}\, t_d^{4/3}$

Technology factor

$C = S \cdot K^{-1/3}\, t_d^{-4/3}$

(9) Development sub-cycle

$t = \frac{1}{2t_{od}^2}$ $\quad t_{od} \rightarrow$ design time

$\boxed{t_{od} = t_d/\sqrt{6}}$

$\boxed{K_d = \frac{y_d(t)}{.95}}$

$\boxed{K_d = K/6}$

$\boxed{M_{od} = D\, t_{od}\, e^{-1/2}}$

$\boxed{\cancel{D_0 = \frac{K_d}{\sqrt{6}\, t_{od}^2}}}$

$\boxed{D = \frac{K_d}{t_{od}^2}}$

\# If "development" word comes it means it is ~~a~~ sub-cycle.

Software Risk Management

→ Risk management means dealing with a concern before it becomes a crisis

→ Risk management is the process of identifying, addressing & eliminating risk before they can damage the project.

RISK

→ It is a problem that could cause some loss or threaten the success of the project, but which has not happened yet.

→ Risk means failure which has not yet occur but which may occur in future.

→

TYPES OF SOFTWARE RISK FACTORS

(1) Dependencies

→ Many risks arise due to dependencies of project on outside agencies or factors.

→ Some typical dependency-related risk factors are:

(a) Availability of trained, experienced people

(b) Inter-group dependencies

(c) Customer-furnished items or information

(d) Internal & external subcontractor relationships.

(2) Requirement issues

→ Wrong product can be build if there is requirement issues

→ Factors are:

- Lack of clear product vision
- Lack of agreement on product requirements
- Unprioritized requirements
- Rapidly changing requirements
- Inadequate impact analysis of requirement changes.

(3) Management issues

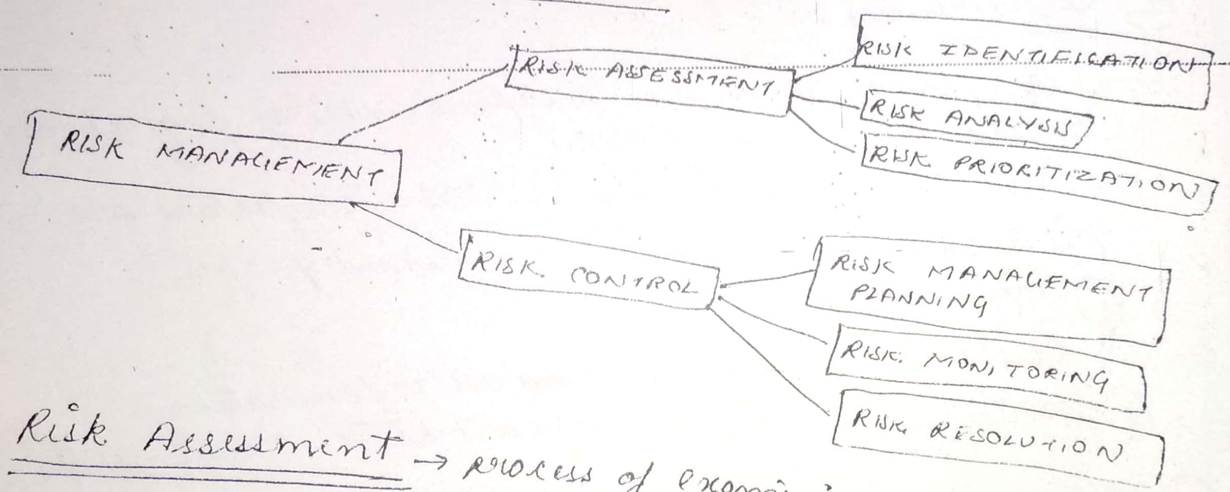
Few Mgmt Issues lead to risks:

→ Inadequate planning & task identification
→ Managers or Customers with unrealistic expectations
→ Staff personality conflicts
→ Poor communication

(4) Other Risk Categories

- Unavailability of adequate testing facilities
- Turnover of essential personnel
- Unachievable performance requirements
- Technical approaches that may not work

RISK MANAGEMENT ACTIVITIES



Risk Assessment → process of examining project & identifying areas of potential risk.

→ It consist of 3 activities:
(a) Identifying the risks
(b) Analyzing them
(c) Assigning priorities to each of them.

(a) **RISK IDENTIFICATION**

→ Risk can be identified with the help of checklist of common risk area of software project
OR
→ Examining contents of organizational database of previously identified risks.

(b) Risk Analysis

→ examining how project outcome might change with modification of risk or if risk occurs.

(c) Risk Prioritization

→ It helps the project focus on its most severe risk by assessing the risk exposure.

→ Exposure is the product of the probability of incurring a loss due to the risk & potential magnitude of that loss.

→ Prioritization can be done in a quantitative way, by estimating the probability (·1 - 1·0) & relative loss on a scale of 1 to 10.

→ Higher the exposure, more aggressively the risk should be tackled.

→ Rating Risk as High, Medium or Low & High rated risk need to be handle first.

RISK - CONTROL

→ Process of managing risks to achieve the desired outcomes.

(a) Risk Management ~~Risk~~ Planning

→ produce a plan for dealing with each significant risk.

→ Record decision in the plan, so that both customer & developer can review how problem are to be avoided as well as how they are to be handled when they arise.

(b) Risk Monitoring

→ Monitor the project as development progresses,

→ periodically re-evaluating the risk, their probability & its impact.

(c) Risk resolution

→ It is the execution of the plans for dealing with each risk.